

Frontend Challenge: X-AI Product Experience Prototype

Thank you for your interest in RacoAI.

We appreciate the time and care you put into exploring this task, which is designed to showcase your approach to frontend engineering, UI/UX design, and interaction design. It reflects how we think about building thoughtful, high-quality product experiences through clarity, intention, and craft. We encourage you to enjoy the process, make deliberate design decisions, and demonstrate how you bring ideas to life through clean implementation and purposeful motion.

Duration: 5 days

“The count down for the deadline starts from the day you received the Email”

Goal

Design and build a **high-fidelity interactive product experience** for Xai that demonstrates:

- UI/UX clarity
- Design-to-code execution
- Advanced motion & 3D interaction
- Engineering discipline and polish

This is **not** a landing page clone.

This is a **product-quality interactive UI** with intentional motion.

High-Level Overview

You will design and implement a **single-page interactive product experience** that visually explains how Xai turns raw data into insights.

The experience should feel:

- Calm but powerful
- Technically confident
- Designed for decision-makers
- Clearly “AI product” — not marketing fluff

Think: **Stripe + Linear + Vercel-level polish**, but with **custom motion & 3D depth**.

Product Concept (What You're Building)

Product Name (use this)

Xai – Intelligence Workspace

Core Narrative

“From raw data → structured intelligence → actionable insight → AI Automations”

The UI should visually *walk the user through this transformation* using animation, geometry, and motion.

Required Sections (Single Page)

1. Hero Section — *Data* → *Intelligence*

- Minimal headline + subtext
- Centerpiece animated visual:
 - Abstract geometry or particles representing “raw data”
 - Smooth transformation into a structured form (grid, nodes, panels)
- Motion must **respond to scroll or cursor**
- Use:
 - **Three.js** (preferred) OR advanced SVG + GSAP
 - **Framer Motion** for UI choreography

⊘ No stock illustrations

⊘ No Lottie files

2. Interactive Insight Flow

A horizontally or vertically animated section that explains **3 stages**:

1. Ingest Data
2. Analyze with AI
3. Generate Insight

Each stage should:

- Animate in/out based on scroll
- Have micro-interactions (hover, focus, transitions)
- Use **geometry-based animation** (lines, masks, transforms)

This is where **GSAP + ScrollTrigger** or **Framer Motion + layout animations** should shine.

3. Intelligence Dashboard Preview

A **mock product UI**, not marketing cards.

Include:

- Sidebar navigation
- Main content panel
- Charts / cards / tables (static data is fine)

Add:

- Subtle entrance animations
- Hover feedback
- State transitions (e.g. switching tabs)

Focus on:

- Visual hierarchy
- Spacing
- Calm, professional typography

4. Signature Interaction (WOW Moment)

One **deliberate, impressive interaction**, for example:

- A 3D object that reacts to scroll
- A geometry morph triggered by user action
- A data cluster that reorganizes itself
- A depth-based parallax system

This should feel:

“This person understands motion, math, and intent.”

Design Requirements (Figma)

You must create:

- A **high-fidelity Figma design**
- Proper use of:
 - Auto Layout
 - Components & variants
 - Consistent spacing system
- Desktop-first (responsive thinking is a plus)

No need to overdesign every screen to focus on **clarity and quality**.

Engineering Requirements (Next.js)

- Next.js (App Router preferred but not required)
- Clean component structure
- Reusable UI components
- Thoughtful animation architecture
- No backend needed (mock data is fine)

Animation stack:

- **Framer Motion** (required)
- **GSAP** (for advanced timelines / scroll)
- **Three.js or React Three Fiber** (at least one meaningful use)

What NOT to Do

- ⊘ No templates
- ⊘ No UI kits
- ⊘ No copied Dribbble shots
- ⊘ No excessive pages
- ⊘ No shallow animations

This is about **taste, restraint, and execution**.

Deliverables

1. Product documentation and Figma File

- Clear articulation of the **product idea and page structure in a pdf/docx file**, demonstrating how you translated the high-level concept into a concrete interface
- Link to the publicly accessible Figma design file
- Well-organized pages and frames
- Proper use of components, variants, and a consistent layout system

This deliverable is intended to help us understand how you decompose a high-level product vision into clear UI structure, user flows, and design decisions.

2. GitHub Repository (Engineering Execution)

- Public repository link
- Clear and concise README including:
 - Project overview and technical approach
 - Technology stack used
 - Instructions to run the project locally
 - A short explanation of key animation and interaction decisions, shared via a publicly accessible Google Drive or YouTube video

3. Live Deployment (Quality & Performance)

- Live URL hosted on Vercel or Netlify
- Smooth interactions and reliable performance across the experience

Evaluation Criteria

We will evaluate based on:

UI / UX

- Visual hierarchy
- Typography & spacing
- Professional restraint

Motion & Interaction

- Purposeful animation
- Smoothness & performance
- Understanding of timing & easing

Engineering Quality

- Code structure
- Component reuse
- Animation architecture

Product Thinking

- Does the UI *explain the product*?
- Is there clarity of intent?